

Data Quality & Documentation

Project: Predictive Maintenance – Equipment Failure Prediction Pipeline

Dataset: NASA CMAPSS Turbofan Engine Dataset

Domain: Manufacturing / Aerospace

Data Type: Time-series sensor data

DEFINE DATA QUALITY STRATEGY:

A data quality strategy is a structured approach to ensure that data is accurate, complete, consistent, timely, and reliable across all pipeline layers (Bronze, Silver, ML, Predictions) so that analytics and machine learning outputs are trustworthy and decision-ready.

Goal:

Ensure sensor, feature, and prediction data are:

- Complete
- Accurate
- Consistent
- Traceable end-to-end

Strategy Table:

Layer	Quality Focus
Bronze	Raw data integrity
Silver	Feature correctness
ML Input	No leakage / nulls
Predictions	Valid probability & labels

Validation Rules Definition:

Validation rules are predefined checks applied at each stage of the data pipeline to ensure data correctness, consistency, and reliability before it is used for analytics or ML.

Bronze Layer (Raw Sensor Data):

Purpose: Ensure ingested data matches source quality.

Rule	Validation
engine_id	NOT NULL
cycle	≥ 1
sensor values	Numeric
row count	> 0
schema	Matches CMAPSS format

Validating Null Values:

▶ 2 minutes ago (3s)3Python

```
dq_bronze = {  
    "null_engine_id": bronze_df.filter(col("engine_id").isNull()).count(),  
    "null_cycle": bronze_df.filter(col("cycle").isNull()).count(),  
    "invalid_cycle": bronze_df.filter(col("cycle") <= 0).count()  
}  
  
dq_bronze
```

> [See performance \(3\)](#)

{'null_engine_id': 0, 'null_cycle': 0, 'invalid_cycle': 0}

Row Count Check:

▶ 2 minutes ago (1s)4Python

```
bronze_df.count()
```

> [See performance \(1\)](#)

20631

Status Logic:

▶ 1 minute ago (<1s)5: Cell 5Python

```
import builtins  
bronze_status = "PASS" if builtins.sum(dq_bronze.values()) == 0 else "FAIL"  
bronze_status
```

'PASS'

Silver Feature Validation:

Purpose: Ensure feature correctness.

Silver Validation Table:

Rule	Validation
Rolling features	NOT NULL
Delta features	Numeric
Cycle continuity	No gaps per engine
Feature range	No abnormal spikes

Outcome:

- Features correctly derived from Bronze data

Null Feature Check:

12:46 PM (20s)3Python

```
feature_cols = [c for c in silver_df.columns if "roll" in c or "delta" in c]

null_features = {
    c: silver_df.filter(col(c).isNull()).count()
    for c in feature_cols
}
null_features
```

> See performance (42)

{'s1_roll_mean': 0,
's1_delta': 0,
's2_roll_mean': 0,
's2_delta': 0,
's3_roll_mean': 0,
's3_delta': 0,
's4_roll_mean': 0,
's4_delta': 0,
's5_roll_mean': 0,
's5_delta': 0,
's6_roll_mean': 0,
's6_delta': 0,
's7_roll_mean': 0,
's7_delta': 0,
's8_roll_mean': 0,
's8_delta': 0,
's9_roll_mean': 0,
's9_delta': 0,
's10_roll_mean': 0,

Cycle Consistency:

12:48 PM (2s)5Python

```
silver_df.limit(5).display()
```

> See performance (1)

engine_id	cycle	1.2 op1	1.2 op2	1.2 op3	1.2 s1_roll_mean	1.2 s1_delta	1.2 s2_roll_mean	1.2 s2_delta
1	1	1	-0.0007	-0.0004	100	518.67	0	641.82
2	1	2	0.0019	-0.0003	100	518.67	0	641.985
3	1	3	-0.0043	0.0003	100	518.67	0	642.1066666666667
4	1	4	0.0007	0	100	518.67	0	642.1675
5	1	5	0.0010	0.0002	100	518.67	0	642.208

5 rows | 1.64s runtime

Refreshed 6 minutes ago

Label Validation:

Label validation ensures that the failure labels used for ML training are correct, consistent, and logically aligned with the Remaining Useful Life (RUL) definition. To verify that the generated failure labels correctly represent imminent engine failure based on RUL thresholds.

Label Validation Rules:

Check	Result
Label values	Only 0 or 1
Threshold applied	$RUL \leq 30$
Temporal correctness	No future leakage

Outcome:

- Failure labels logically aligned with engine degradation

Validate Labels:

▶ 1 minute ago (1s)3Python

```
labels_df.groupby("failure_label").count().show()
```

> [See performance \(1\)](#)

```
+-----+
|failure_label|count|
+-----+
|           0|17531|
|           1| 3100|
+-----+
```

Threshold Check:

▶ 1 minute ago (1s)4

```
labels_df.filter(
  (col("RUL") <= 30) & (col("failure_label") != 1)
).count()
```

> [See performance \(1\)](#)

0

ML Input Validation:

ML Input Validation ensures that the dataset fed into the machine learning model is clean, consistent, complete, and aligned with what the model was trained on.

ML Dataset Rules:

Rule	Validation
Features vector	No NULL
Label	Binary (0 / 1)
Train/Test split	Engine-level separation
Class distribution	Acceptable imbalance

Model Input Summary

- Model: Spark ML (Logistic Regression)
- Input: Silver features + failure label
- Output: Probability & prediction

Outcome:

- ML inputs validated and safe for training

Class Distribution:

```
▶ Just now (1s) 3 Python
train_df.groupBy("failure_label").count().show()

> See performance \(1\)

+-----+
|failure_label|count|
+-----+
|           0|17531|
|           1| 3100|
+-----+
```

Prediction Output Validation:

Prediction Output Validation ensures that the machine learning model's predictions are correct, interpretable, reliable, and usable for business decisions such as maintenance planning and risk monitoring.

Prediction Schema:

Column	Description
probability	Failure probability (0–1)
prediction	0 = Safe, 1 = Risk
risk_level	LOW / MEDIUM / HIGH

Validation Rules:

Rule	Status
Probability range	$0 \leq p \leq 1$
Prediction values	0 and 1
Risk mapping	Correct

Probability Check:

▶ 02:44 PM (1s) 3

```
pred_df.filter(
  (col("failure_probability") < 0) | (col("failure_probability") > 1)
).count()
```

> [See performance \(1\)](#)

0

Risk Level Validation:

▶ 02:45 PM (1s) 4

```
pred_df.groupBy("risk_level").count().show()
```

> [See performance \(1\)](#)

+-----+
|risk_level|count|
+-----+
LOW	12661
MEDIUM	117
HIGH	218
+-----+

Audit & Control Table:

An Audit & Control Table is a centralized table used to track, monitor, and validate data processing activities across the entire pipeline (Bronze → Silver → ML → Predictions).

Purpose:

- Monitor pipeline health and job execution
- Track record counts and data quality checks
- Detect failures, anomalies, or data loss
- Support troubleshooting, compliance, and client audits

Table:

Column	Description
layer_name	Bronze / Silver / ML
record_count	20637
validation_status	PASS
validation_date	22/01/26

Business Value:

Provides end-to-end visibility and confidence that the predictive maintenance pipeline is producing reliable and auditable results.